

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Formative Assessment 4

Formative Assessment (FA-4)

Class: Class 10

Assessment Type: Formative Assessment (FA-4)

Total Marks: 30

Duration: 90 minutes

Topics Covered: SQL Database Management

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which SQL command is used to retrieve data from a database?
 - b. GET
 - c. RETRIEVE
 - d. SELECT
 - e. FETCH
- b. The WHERE clause in SQL is used with:
 - c. SELECT only
 - d. SELECT, UPDATE, DELETE
 - e. INSERT only
 - f. CREATE only
- c. Which aggregate function returns the total sum of a numeric column?
 - d. COUNT()
 - e. SUM()
 - f. TOTAL()
 - g. ADD()
- d. Which SQL command is used to remove a table completely from the database?
 - e. DELETE TABLE
 - f. REMOVE TABLE
 - g. DROP TABLE
 - h. TRUNCATE TABLE
- e. The GROUP BY clause is used to:
 - f. Sort data
 - g. Filter rows
 - h. Group rows with same values for aggregate functions
 - i. Delete duplicates

Section II: Short Answer (5 x 2 = 10 marks)

- a. Write SQL queries for a table Students(roll_no, name, class, section, marks):
- Find all students of class '10' with marks above 80
 - Count the number of students in each section
 - Find the maximum and minimum marks
 - List students in alphabetical order of names
- b. Differentiate between WHERE and HAVING clause with examples.
- c. What is a **primary key**? Can a table have more than one primary key? Explain.
- d. Write the output of:

```
SELECT UPPER('hello world');  
SELECT LENGTH('Karnataka');  
SELECT CONCAT('ICT', ' Class', ' 10');
```

- e. Explain the ACID properties of a transaction with one example each.

Section III: Long Answer (5 x 1 = 5 marks)

- a. Write SQL queries for a **Library Management System** with tables:
- Books(book_id, title, author, price, quantity)
 - Members(member_id, name, phone, membership_date)
 - Issue(issue_id, book_id, member_id, issue_date, return_date)
- (a) List all books with price above 500 sorted by title (b) Find members who have not returned any book (c) Count how many times each book has been issued (d) Find the total revenue from all books (e) Display members who joined in the last 6 months

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Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Connects to a SQLite database
- Creates tables for a Student Management System
- Inserts at least 10 records
- Performs all CRUD operations (Create, Read, Update, Delete)
- Generates a report with aggregate functions (COUNT, AVG, MAX, MIN)
- Handles database exceptions properly

Part C: Notebook Maintenance (5 marks)

- a. Neatness and organisation of notebook (2 marks)
- b. Completion of all assigned classwork and homework (2 marks)
- c. Proper headings, dates, and indexing (1 mark)